Fundamentals 19

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1. A necessary condition for the series to converge is that its terms go to 0 as $n \to \infty$. But if $|x| \geq 1$ then x^n diverges, so the series diverges.

2. By the Cauchy condensation test, $\sum_{k=2}^{\infty} \frac{1}{k \log(k)^\alpha}$ converges iff $\sum_{k=2}^{\infty} \frac{2^k}{2^k \log(2^k)^\alpha}$ converges. We see that

$$\sum_{k=2}^{\infty} \frac{2^k}{2^k \log(2^k)^\alpha} = \sum_{k=2}^{\infty} \frac{1}{\left(\frac{\log_2(2^k)}{\log_2(e)}\right)^\alpha} = \sum_{k=2}^{\infty} \frac{\log_2(e)^\alpha}{\log_2(2^k)^\alpha} = \log_2(e)^\alpha \sum_{k=2}^{\infty} k^{-\alpha}.$$

This is a p-series, so it converges iff $\alpha > 1$.

3. $n^{1/\log(n)} = e$.

4. Define

$$S_n = \sum_{k=1}^{n} (-1)^k a_k.$$

Since a_n is decreasing, we know that

$$S_{2n} \leq S_{2n-2} \leq S_{2n-4} \leq \cdots \quad \text{and} \quad S_{2n-1} \geq S_{2n-3} \geq S_{2n-5} \geq \cdots,$$

so the even indexed partial sums form a decreasing sequence and the odd indexed partial sums form an increasing sequence. Also,

$$S_{2n} - S_{2n-1} = \sum_{k=1}^{2n} (-1)^k a_k - \sum_{k=1}^{2n-1} (-1)^k a_k = a_{2n}.$$

Since $a_n \to 0$, the difference between the two partial sums converges to 0, so they converge to the same thing. Since they are both monotonous, S_n converges.

5. Since $\lim_{n\to\infty} |a_n| = 0$, $\lim_{n\to\infty} -|a_n| = 0$. Also, since $-|a_n| \leq a_n \leq |a_n|$, we have that

$$\lim_{n\to\infty} -|a_n| \leq \lim_{n\to\infty} a_n \leq \lim_{n\to\infty} |a_n|,$$

so $\lim_{n\to\infty} a_n = 0$.

6. We have that $\sum_{k=1}^{\infty} a_k$ converges but $\sum_{k=1}^{\infty} |a_k|$ doesn't. For contradiction, assume $\sum_{a_k \geq 0} a_k = L$ for some $L \in \mathbb{R}$. We have that

$$\sum_{k=1}^{\infty} a_k = \sum_{a_k \geq 0} a_k + \sum_{a_k < 0} a_k = L + \sum_{a_k < 0} a_k.$$

But $\sum_{k=1}^{\infty} a_k$ converges, so $\sum_{a_k<0} a_k$ must also converge. But that would mean $\sum_{k=1}^{\infty} |a_k|$ converges, which is a contradiction. So $\sum_{a_k \geq 0} a_k$ diverges, and all terms are positive, it must go to ∞. Similarly, $\sum_{a_k<0} a_k \to -\infty$.

7. I tried proving it this way:

Assume $L \geq 0$ and define ϕ iteratively as:

$$\phi(1) = \text{the first k such that } a_k \geq 0,$$

$$\phi(n) = \begin{cases} \text{the first } k \text{ such that } a_k \geq 0 \text{ and } k \geq \phi(n-1) & \text{if } \sum_{k=1}^{n-1} a_{\phi(k)} \geq L, \\ \text{the first } k \text{ such that } a_k < 0 \text{ and } k \geq \phi(n-1) & \text{if } \sum_{k=1}^{n-1} a_{\phi(k)} < L. \end{cases}$$

This is well-defined by the previous problem. We wish to show that $\sum_{k=1}^{\infty} a_{\phi(k)} = L$.

Define indices k_n as

$$k_0 = 1,$$

$k_n = \text{the first } k > k_{n-1} \text{ such that } a_{\phi(k)} \text{ and } a_{\phi(k_{n-1})} \text{ have different parity.}$

Then, define the sequence

$$b_n = \sum_{i=k_{n-1}}^{k_n - 1} a_{\phi(i)}.$$

By construction, $b_1 > L$, $b_1 + b_2 < L$, $b_1 + b_2 + b_3 > L$, and so on. Also, $b_{2n-1} \geq 0$ and $b_{2n} < 0$ for all $n \in \mathbb{N}$.

From here it would suffice to show that $|b_n|$ is decreasing and goes to 0, since $\sum_{k=1}^{n} b_k$ is an alternating series. I couldn't figure out how to do that though.

8. We know that there exists a sequence M_n such that $|f_n(x)| \leq M_n$ for all n, and that $\sum_{k=1}^{\infty} M_k$ converges. Now, define

$$g_n(x) = \sum_{k=1}^{n} f_k(x).$$

By Weierstrass M-test, g_n converges uniformly to $\sum_{k=1}^{\infty} f_k$. Then, by the Uniform limit theorem, $\sum_{k=1}^{\infty} f_k$ is uniformly continuous.

9. We can show that $f_n(x)$ converges uniformly to 0 on $[-10, 10]$. Fix $\varepsilon > 0$ and let $N > \frac{10}{\varepsilon}$. Since $x \in [-10, 10]$, $|x| \leq 10$. So if $n > N$ then

$$\left|\frac{x}{n}\right| \leq \frac{10}{n} < \frac{10}{N} < \varepsilon.$$

10. Since $|x^n| \leq 1$ for all $x \in (-1, 1)$ and $n \in \mathbb{N}$, we can let $M_n = 1$, and $\lim_{n\to\infty} M_n = 1$. By Weierstrass M-test, $\sum_{k=1}^{\infty} x^k$ converges uniformly and absolutely on X.

The interval $(-1, 1)$ is the largest interval the series converges uniformly on because for any x with $|x| \geq 1$, $\sum_{k=1}^{\infty} x^k$ diverges.

11. Let $X \in \mathbb{R}$ be any compact set. For all $n \in \mathbb{N}$, define

$$M_n = \sup\left(\left\{\left|\sum_{k=0}^{n} \frac{x^k}{k!}\right| : x \in X\right\}\right),$$

so that $\left|\sum_{k=0}^{n} \frac{x^k}{k!}\right| \leq M_n$. We have that

$$\lim_{n\to\infty} M_n = \sup\left(\left\{\lim_{n\to\infty}\left|\sum_{k=0}^{n} \frac{x^k}{k!}\right| : x \in X\right\}\right) = \sup(\{e^x : x \in X\}),$$

which, by EVT, exists because X is compact (closed and bounded). By Weierstrass M-test, the series converges absolutely and uniformly on X.

12. The Taylor series of $\log(1 + x)$ around 0 is

$$\sum_{k=1}^{\infty} \frac{(-1)^{n-1}x^n}{n}.$$

Because of the x^n term, the series only converges if $x \in (-1, 1)$, so any compact subset of $(-1, 1)$ will work. But the series will not uniformly converge on a compact set that isn't a subset of $(-1, 1)$.

13. We have that

$$\sum_{k=1}^{\infty} a_k \sin(kx) \leq \sum_{k=1}^{\infty} |a_k|.$$

Since $\sum_{k=1}^{\infty} |a_k|$ converges, there exists a convergent sequence M_k such that $|a_k \sin(kx)| \leq |a_k| \leq M_k$ for all k. So by Weierstrass M-test, $\sum_{k=1}^{\infty} a_k \sin(kx)$ converges uniformly on $\mathbb{R}$.